

A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

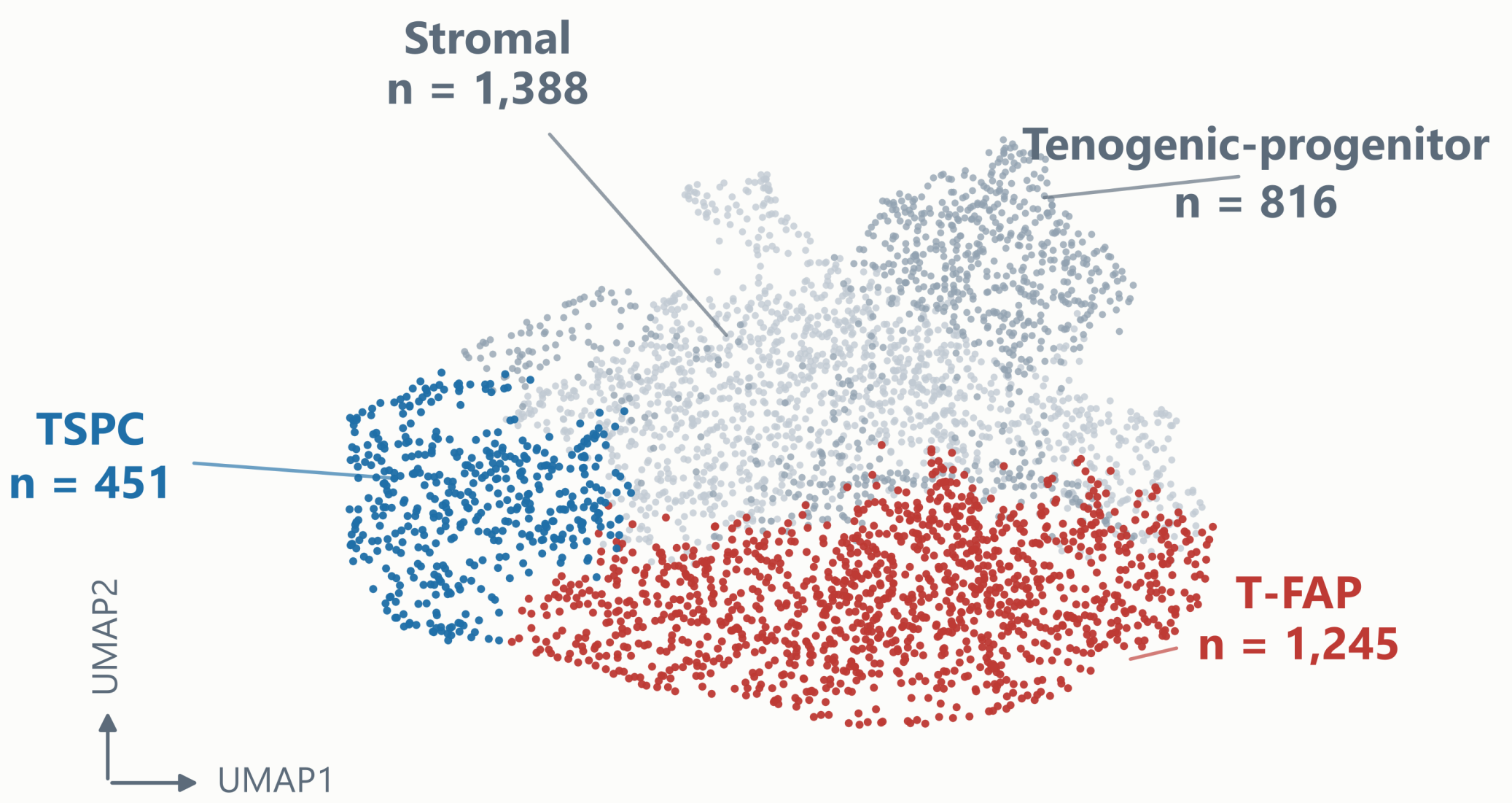
Mapping the transcription-factor program behind the fate decision in tendon progenitors — and compressing it into a validated 23-gene Healing Index.

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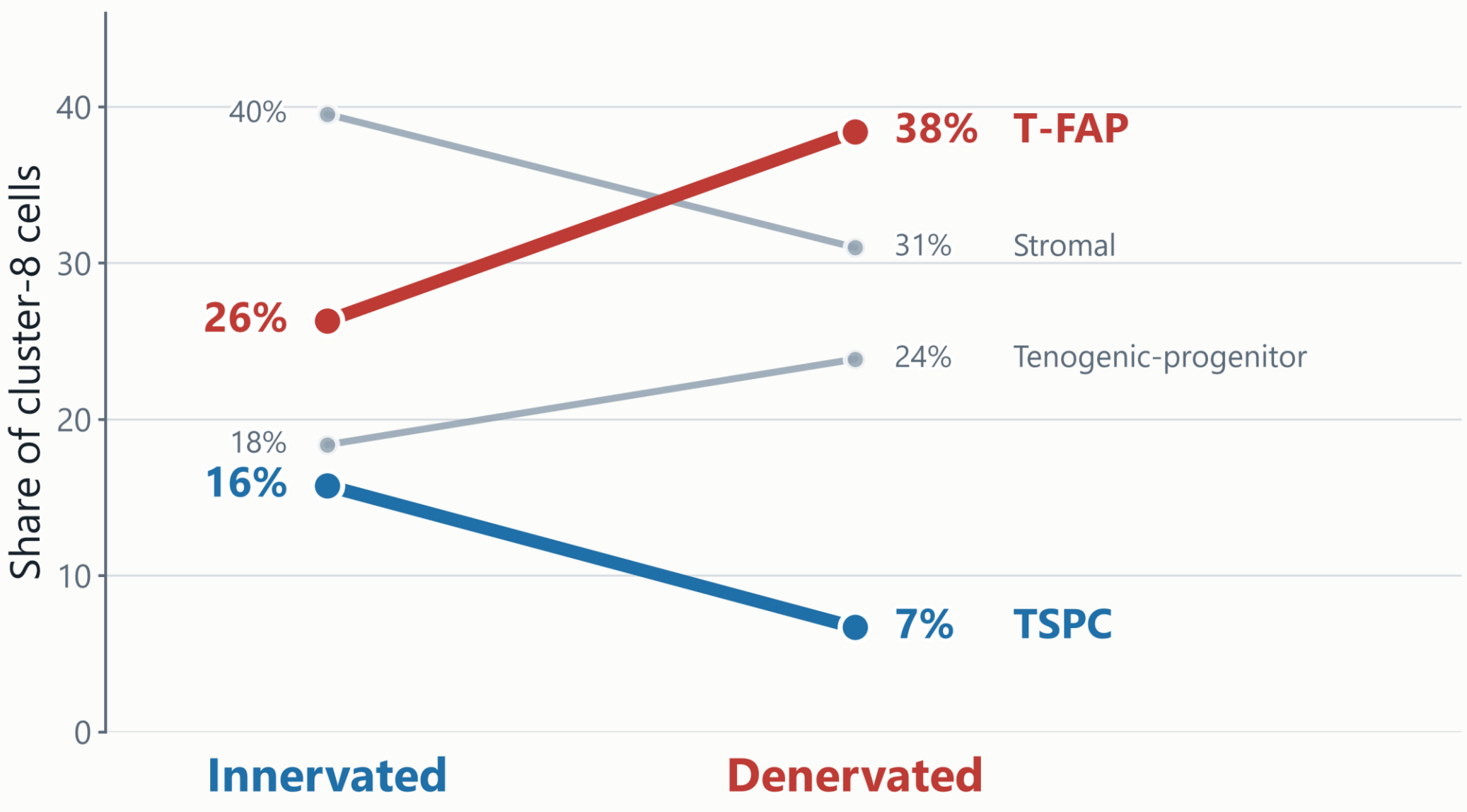
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1 Introduction

Tendon heals badly: repair tissue is scar, not aligned tendon. Two progenitor populations inside the same PDGFRα⁺ pool decide which one forms — tendon stem/progenitor cells (TSPC) rebuild tendon, tendon fibro-adipogenic progenitors (T-FAP) lay down scar (Harvey et al., 2019).



3,900 PDGFRα⁺ cells from injured mouse tendon.
Sub-clustering resolves four populations (Cherief et al., 2023; day 14 post-injury).



Blocking sensory-nerve signalling shifts the pool.
TSPC share falls by more than half; T-FAP becomes the largest population. Nerve signalling is blocked chemical-genetically (TrkA-F592A mice + 1NMPP1), not surgically — “denervated” is shorthand throughout.

OBJECTIVE

Identify the transcription-factor program that carries out this innervation-dependent fate decision, then compress it into a minimal, validated gene signature that scores how regeneratively a tendon is healing.

4 Conclusions

A mechanism-derived 23-gene Healing Index now tracks the regenerative-to-fibrotic shift across two independent tendon datasets.

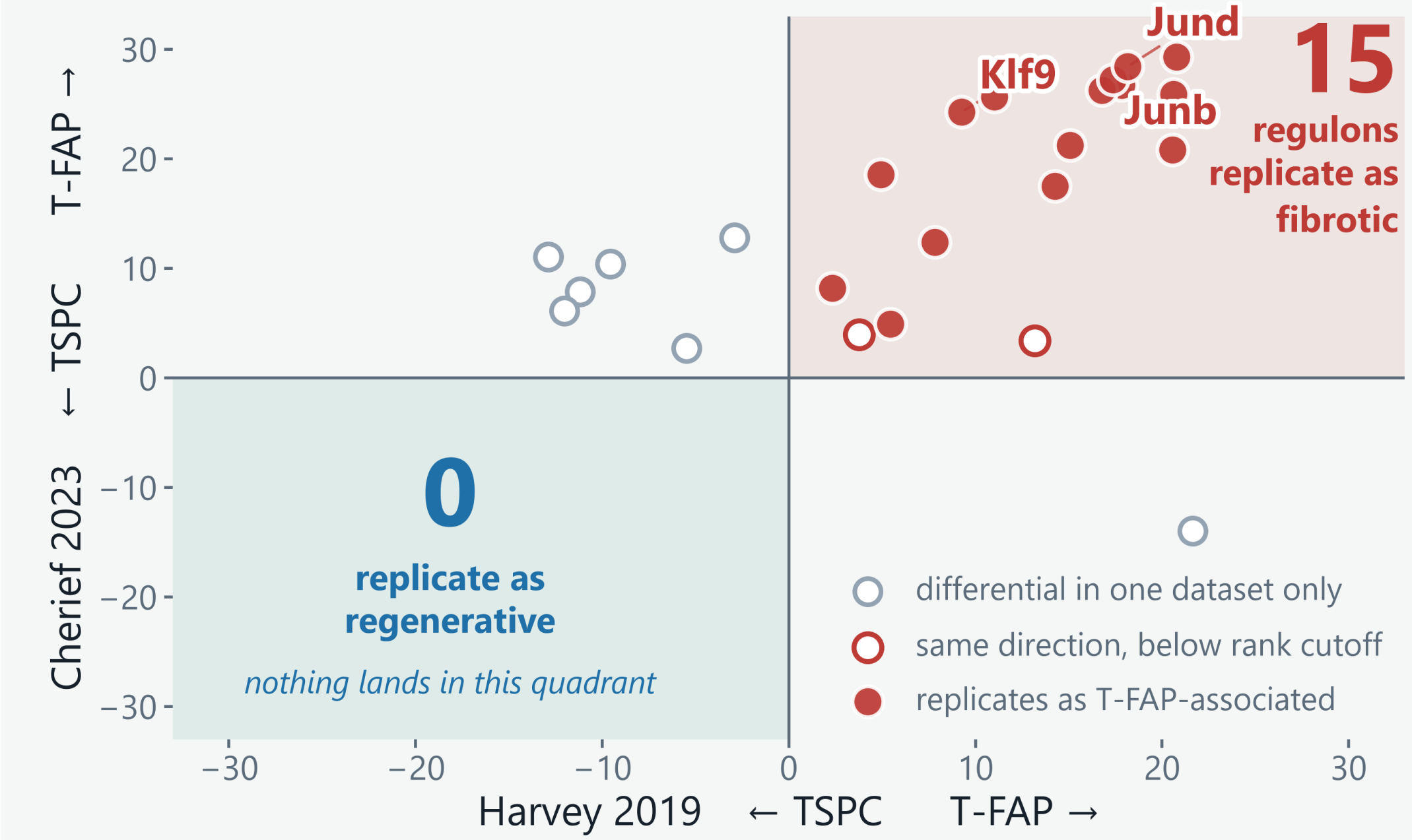
Data & code availability
All data are publicly available (GEO GSE244921; SRA PRJNA506218). No unpublished or confidential data were used. Computation was performed on the Digital Research Alliance of Canada (NIBI). Code and analysis notebooks: github.com/lifangy6/sunnybrook-tendon-fate-mapping

References
1. Cherief, M., et al. (2023). TrkA-mediated sensory innervation of injured mouse tendon supports tendon sheath progenitor cell expansion and tendon repair. Science Translational Medicine, 15(727), eade4619.
4. Van de Sande, B., et al. (2020). A scalable SCENIC workflow for single-cell gene regulatory network analysis. Nature Protocols, 15, 2247–2276.

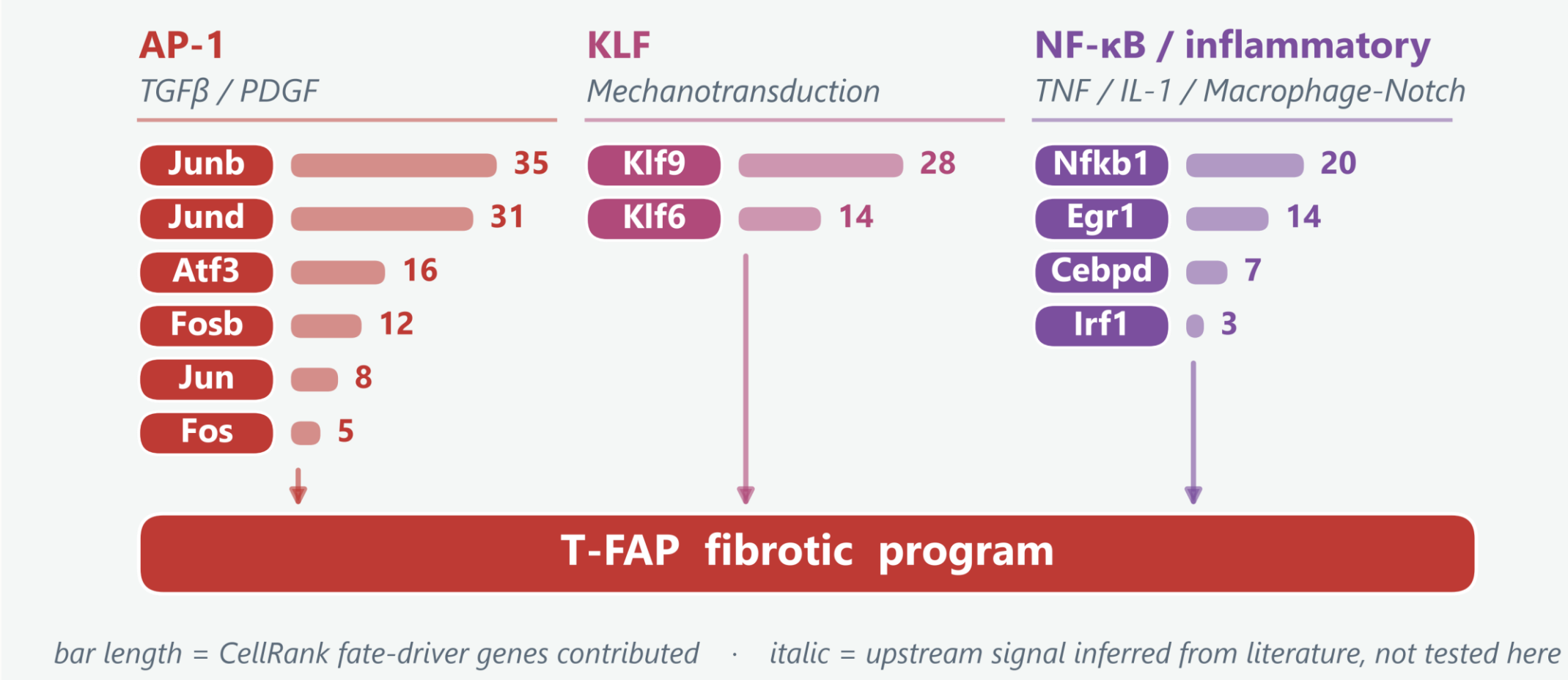
2. Dimitrov, D., et al. (2022). Comparison of methods and resources for cell-cell communication inference from single-cell RNA-Seq data. Nature Communications, 13, 3224.
5. Weiler, P., Lange, M., Klein, M., Pe'er, D., & Theis, F. J. (2024). CellRank 2: Unified fate mapping in multiview single-cell data. Nature Methods, 21, 1196–1205.

3 Results

Which transcription-factor regulons separate the two fates, and do they replicate in a second, independent dataset?



Fifteen regulons replicate as fibrotic. None replicate as regenerative.
A 200-iteration power-matched audit reproduced the empty quadrant, so it is not a sample-size artefact.

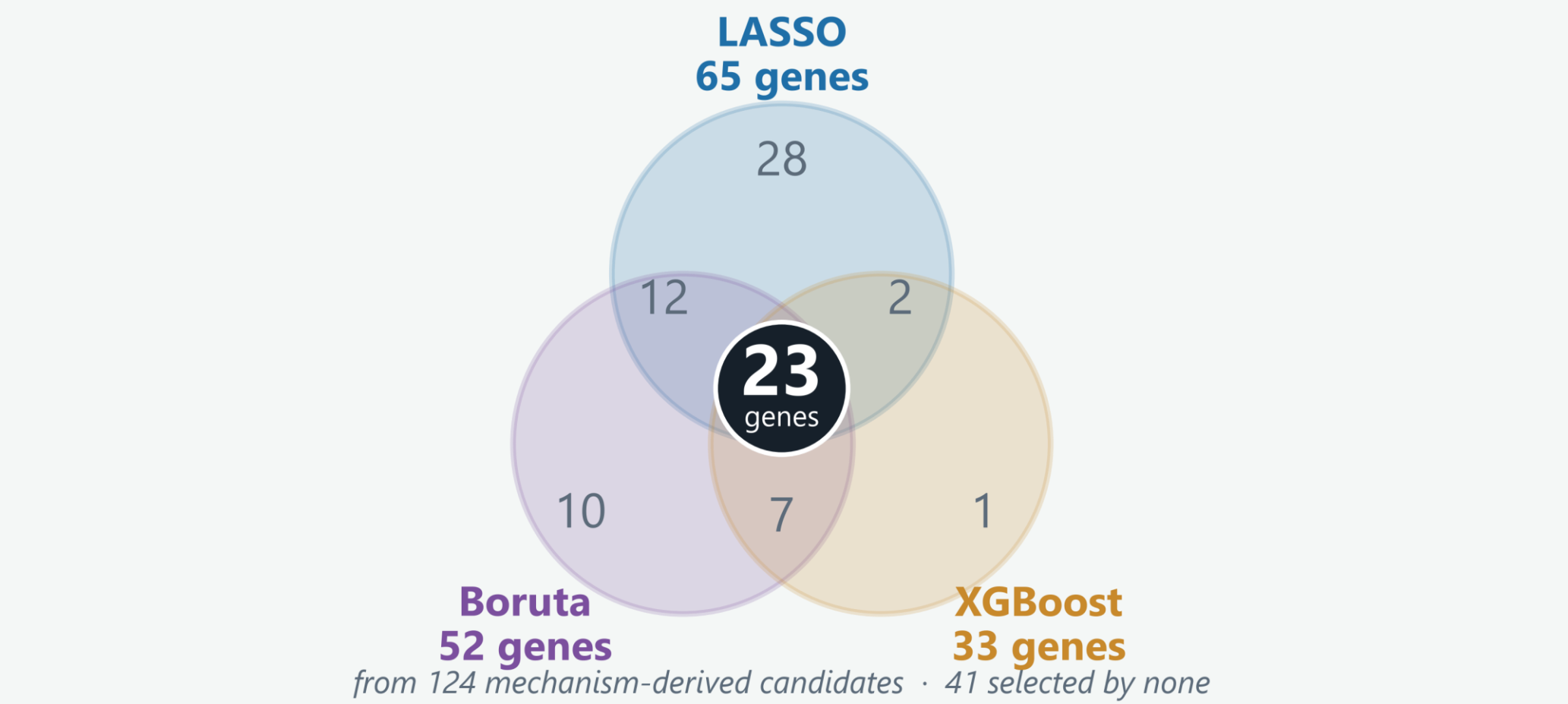


The surviving program is an AP-1 / KLF / NF-κB network.
Junb, Jund and Klf9 contribute the most fate-driver genes; 14 of the 23 final panel genes are their targets. 12 of the 15 regulons are shown — three contribute none.

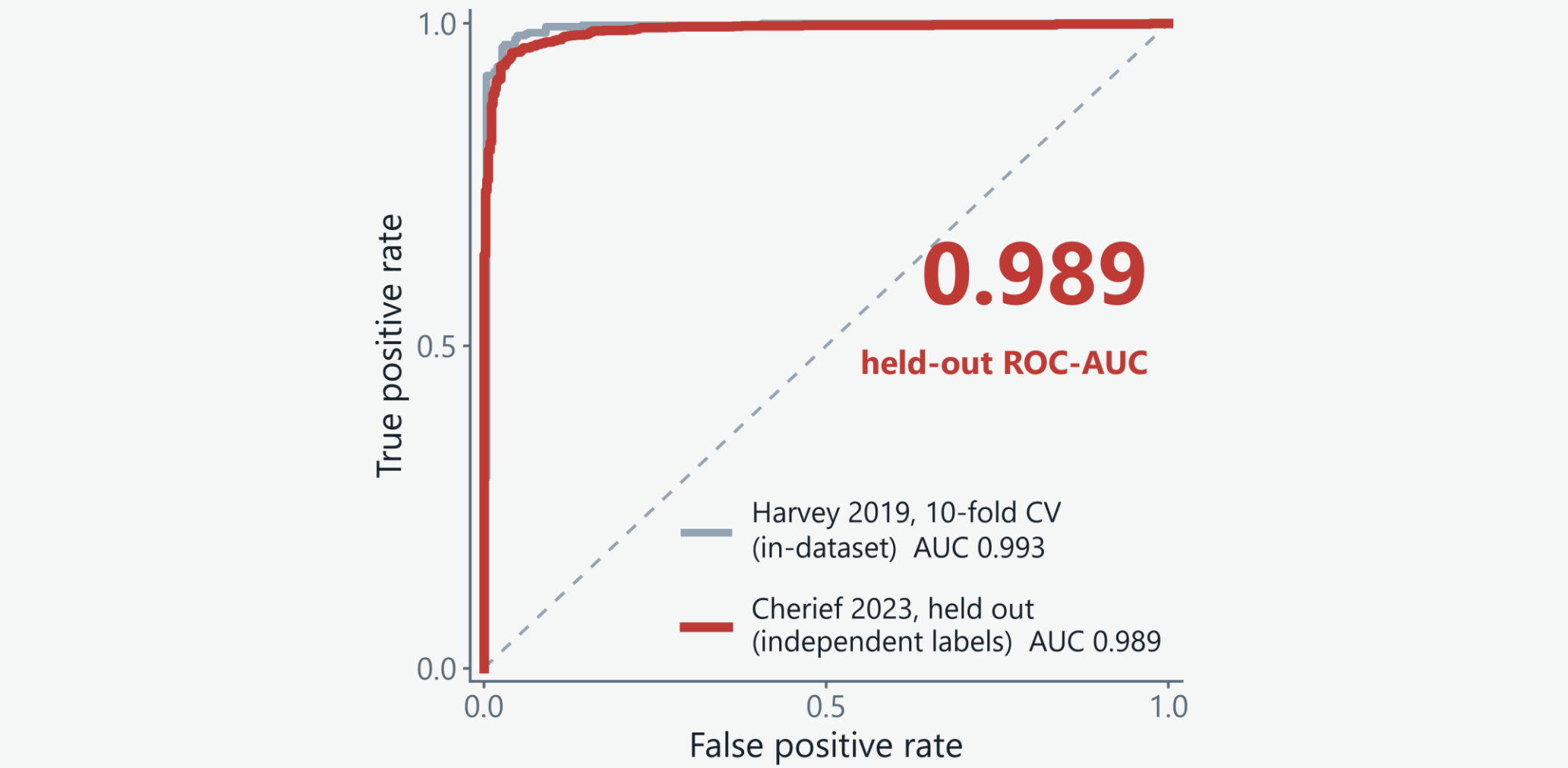
KEY FINDING

The fate decision is asymmetric. Innervation appears to suppress a default fibrotic program rather than to activate a regenerative one — the mirror image of our starting hypothesis.

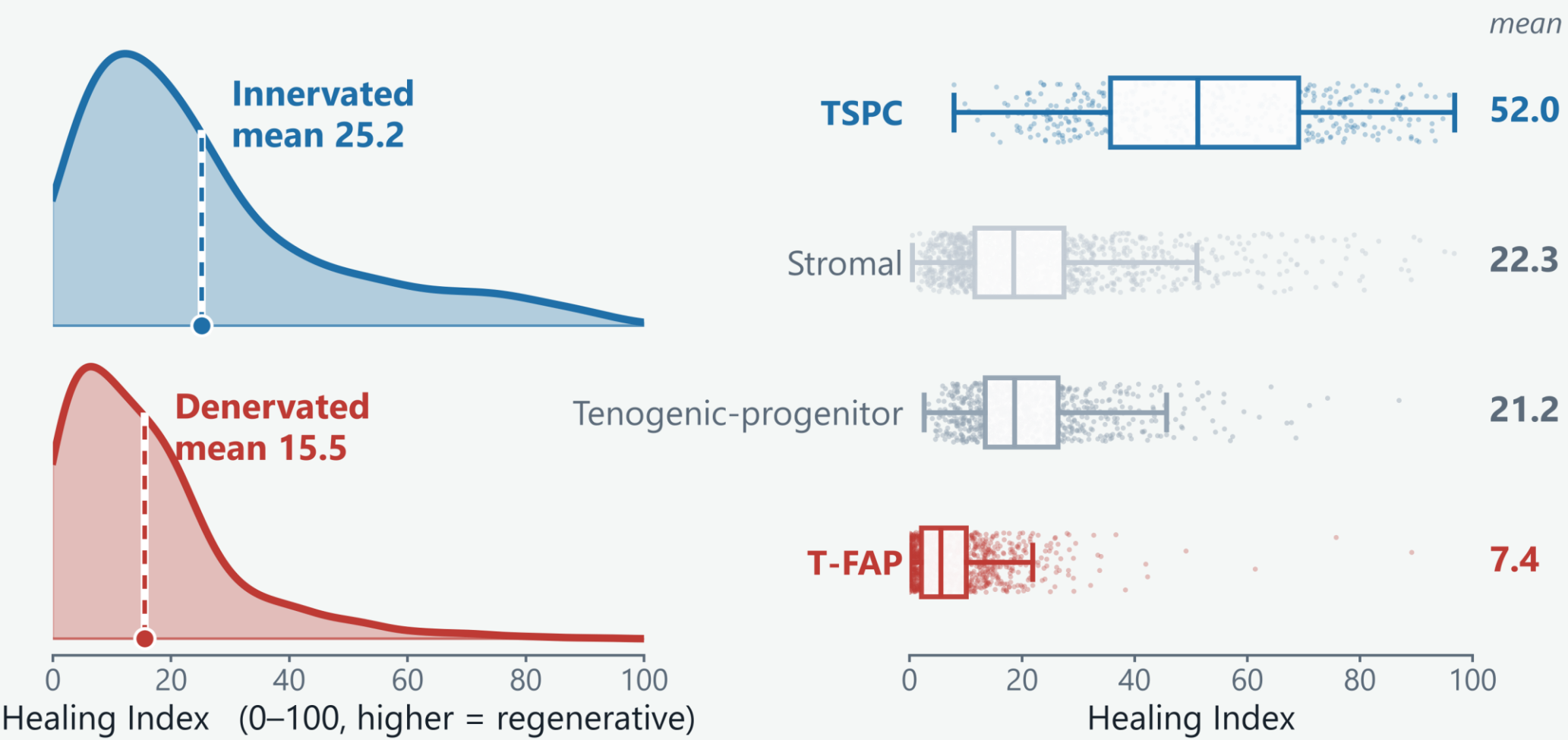
From mechanism to a gene signature



Three methods agree on 23 genes.
LASSO, Boruta and XGBoost ran independently on 124 mechanism-derived candidates.



The panel generalises to a held-out dataset.
Trained on Harvey et al. (2019), then applied unchanged to Cherief et al. (2023) — independently labelled, never seen during selection or training.



The Healing Index tracks innervation status.
It also places two cell identities it never saw in training (Stromal, Tenogenic-progenitor) between the two fates.

LIMITATIONS · WHAT'S NEXT
• No reproducible regenerative regulon is not proof none exists — both datasets are small.
• Neither dataset has biological replicates — validation is cross-dataset, not cross-animal.
• TSPC recall is 77–78% vs. 98–99% for T-FAP: the index flags fibrotic drift better than it confirms repair.
• Next: score newly generated lab tendon scRNA-seq — the first true biological-replicate test.